In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

In [7]:
```
data = pd.read_csv('Body_weight.csv')
```

In [9]:
```
data.head()
```

Out[9]:

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male

In [11]:
```
data.shape
```

Out[11]:
```
(10, 4)
```

In [13]:
```
data.isnull().sum()
```

Out[13]:
```
Weight    0
Height    0
Age       0
Gender    0
dtype: int64
```

In [15]:
```
data['Gender'] = data['Gender'].map({'Male':1,'Female':0})
```

In [17]:
```
x = data.iloc[:,1:4].values
y = data.iloc[:,0:1].values
```

In [19]:
```
x_train, x_test, y_train, y_test = train_test_split(x,y,
        test_size=0.2, random_state = 42)
```

In [21]:
```
model = LinearRegression()
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
```

In [23]:
```
print(y_pred)
print(y_test)
```

```
[[56.85651025]
 [79.10576115]]
[[64]
 [69]]
```

In [29]:
```
print(model.intercept_)
```

```
[-3.05521724]
```

In [31]:
```
print(model.coef_)
```

```
[[32.8353363   0.31729474 14.62311848]]
```

In [35]:
```
from sklearn.metrics import r2_score, mean_squared_error
```

In [37]:
```
print("R2 Score :", r2_score(y_test, y_pred))
```

```
R2 Score : -11.252468343750815
```

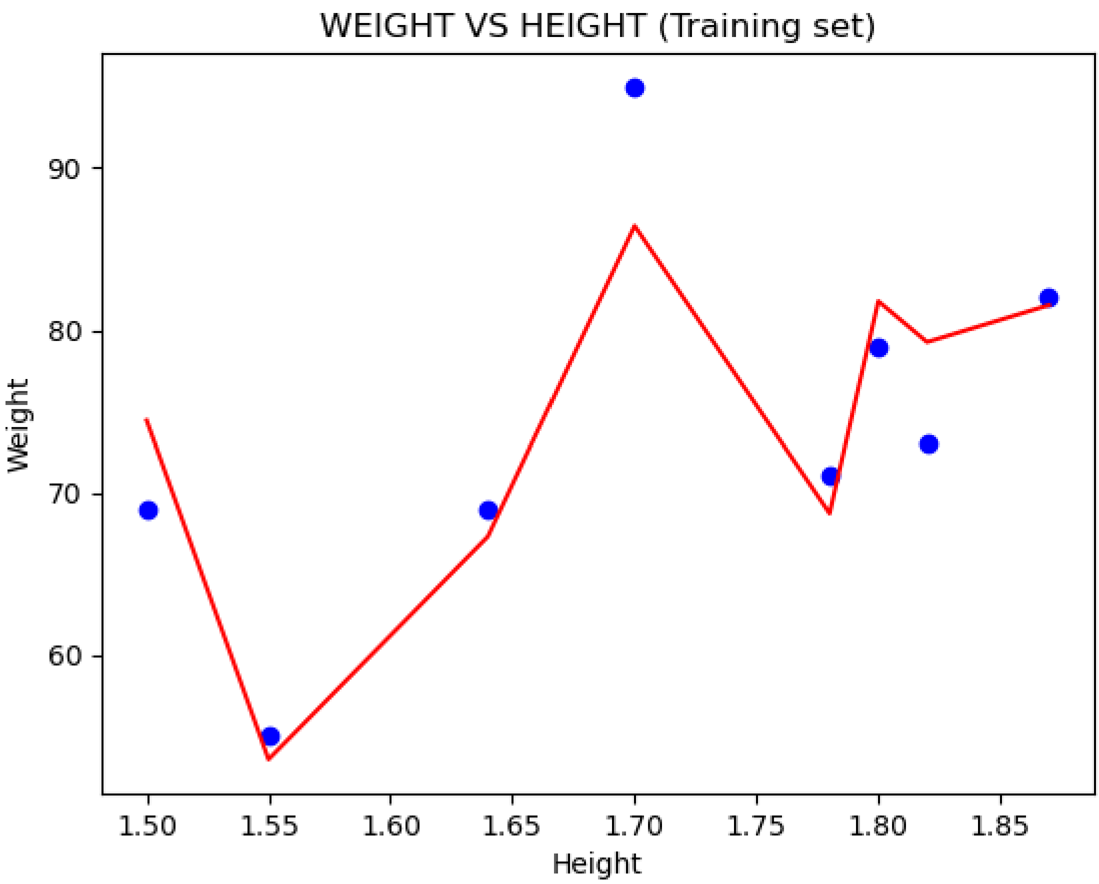
In [39]:
```
print("Mean Squared Error :", mean_squared_error(y_test, y_pred))
```

```
Mean Squared Error : 76.5779271484426
```

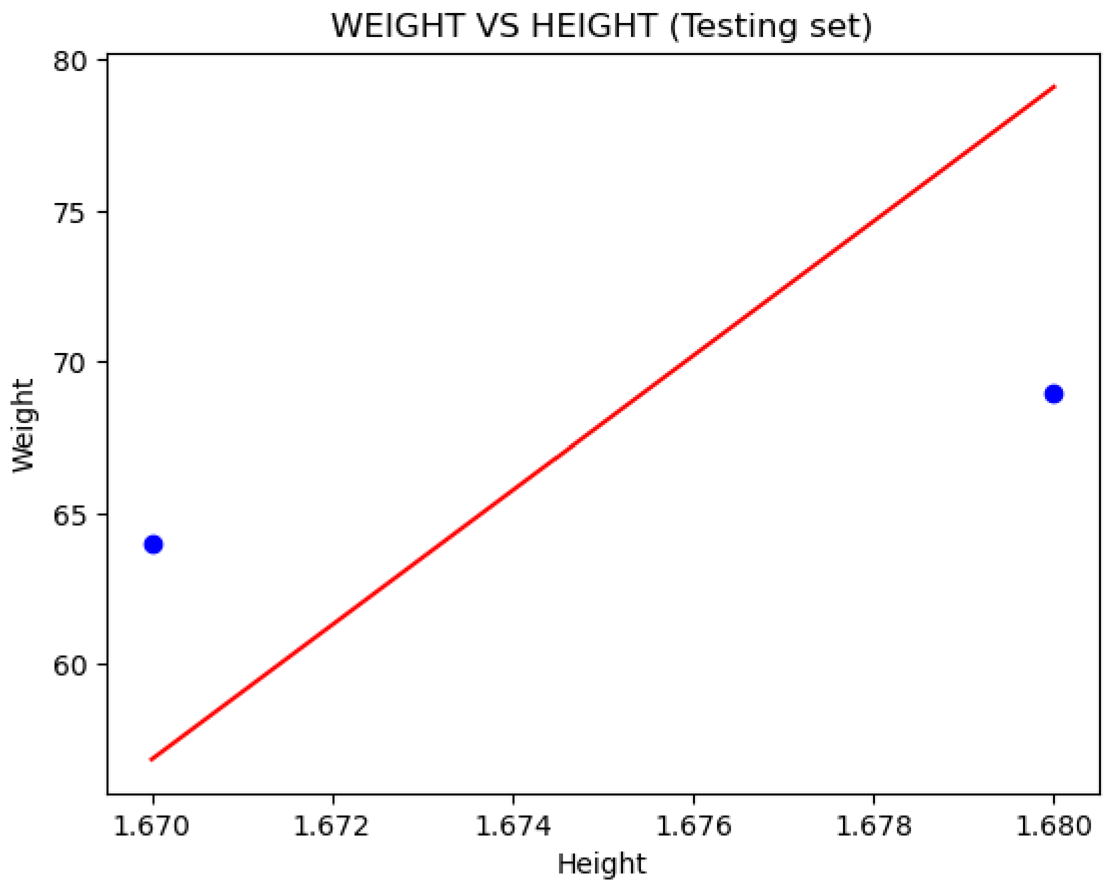
In [41]:
```
new_person = [[1.75,30,1]]
prediction = model.predict(new_person)
print("Predicted Weight =", prediction)
```

```
Predicted Weight = [[78.54858202]]
```

In [43]:
```
import numpy as np

idx = np.argsort(x_train[:,0])

plt.scatter(x_train[:,0], y_train, color='blue')
plt.plot(x_train[idx,0], model.predict(x_train)[idx], color='red')

plt.title('WEIGHT VS HEIGHT (Training set)')
plt.xlabel('Height')
plt.ylabel('Weight')
plt.show()
```



In [45]:
```
idx = np.argsort(x_test[:,0])

plt.scatter(x_test[:,0], y_test, color='blue')
plt.plot(x_test[idx,0], model.predict(x_test)[idx], color='red')

plt.title('WEIGHT VS HEIGHT (Testing set)')
plt.xlabel('Height')
plt.ylabel('Weight')
plt.show()
```



In [ ]: